

Nolan_Week_Two

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Assignment Details

- 1.Bar graph showing product and the sum of sales for each.
- 2.Pie chart showing sum of sales by region.
- 3.Box plots for average sales by region and average quantities by region.
- 4.Histogram of the count of orders by product.
- 5.A radial plot showing total sales (for all products) by month.

Import data from Github

```
url <- "https://raw.githubusercontent.com/bellevue-university/dsc520/master/assignments/version_two/10_
data <- read.csv(url)
## printing out data to validate it matches what is in github
head(data, n = 10)
```

```
##      OrderID  OrderDate Region  Product Quantity UnitPrice TotalSales
## 1           1  1/12/2024  South Product_D         7      34.37    60.14750
## 2           2  8/27/2025  South Product_A        15       8.59    30.07375
## 3           3   5/8/2024  North Product_A        11       2.00    33.98000
## 4           4   4/6/2024   West Product_A         8       3.09    23.39500
## 5           5   9/7/2024  North Product_D         7      24.05   707.18000
## 6           6 10/23/2023  South Product_B        19      74.44   123.20000
## 7           7   8/6/2025   East Product_A        11       6.48    37.22000
## 8           8  4/29/2023  South Product_D        11      12.44   635.40000
## 9           9  4/25/2025  North Product_D         4      29.98   226.67500
## 10          10   5/3/2025   West Product_C         8      30.80   106.11500
```

```
dim(data) # rows and columns
```

```
## [1] 495    7
```

```
summary(data) #basic stats
```

```
##      OrderID      OrderDate      Region      Product
## Min.   : 1.0   Length:495      Length:495      Length:495
## 1st Qu.:127.5   Class :character  Class :character  Class :character
## Median :251.0   Mode  :character  Mode  :character  Mode  :character
## Mean   :251.3
```

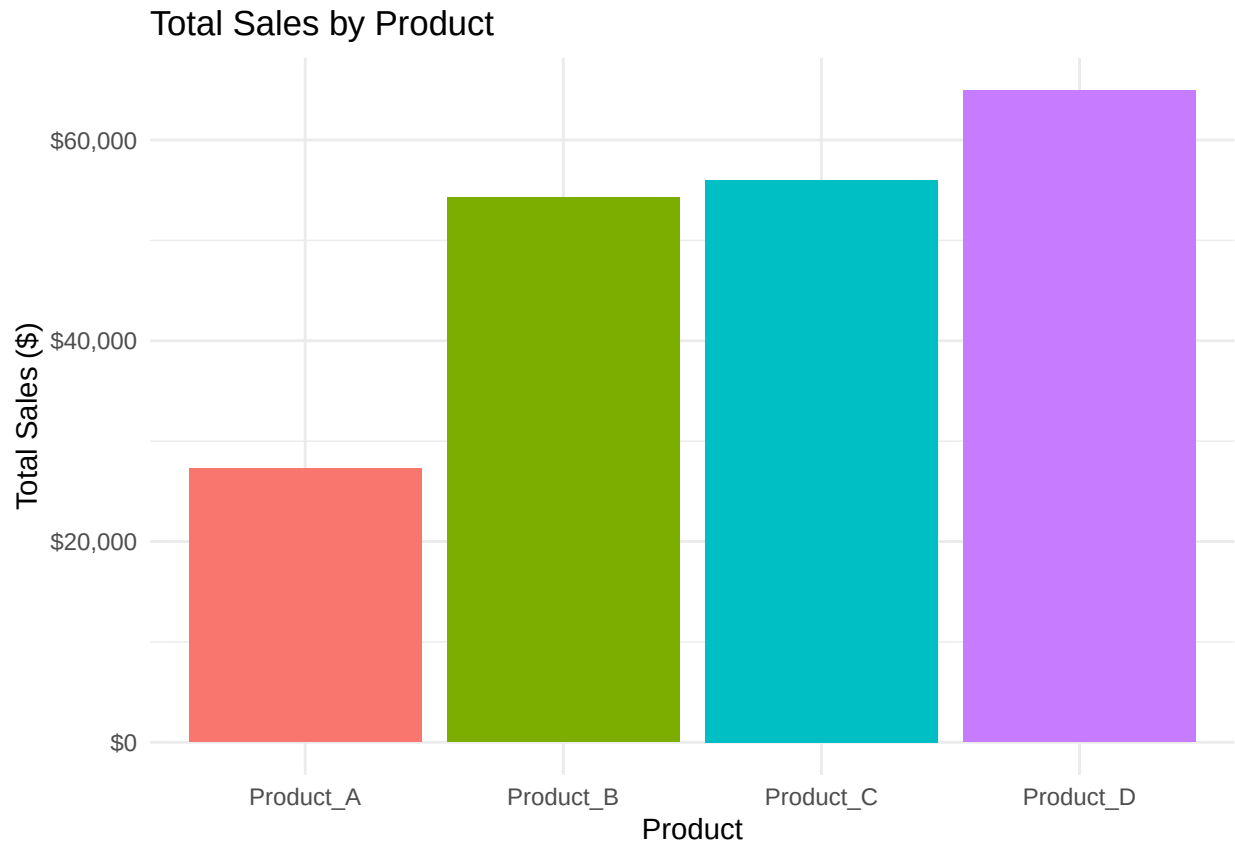
```
## 3rd Qu.:376.5
## Max.    :501.0
##      Quantity      UnitPrice      TotalSales
## Min.    : 1.000    Min.      : 0.98    Min.      : 8.545
## 1st Qu.: 4.000    1st Qu.: 26.74    1st Qu.: 99.905
## Median : 9.000    Median : 50.97    Median : 271.145
## Mean    : 9.592    Mean    : 52.41    Mean     : 409.053
## 3rd Qu.:15.000    3rd Qu.: 77.19    3rd Qu.: 602.115
## Max.    :19.000    Max.     :144.08    Max.     :1895.060
```

```
# data prep. get months organized (MDY format)
data$OrderDate <- mdy(data$OrderDate) # convert to date
data$Month <- month(data$OrderDate) # numeric month
data$Month_Label <- month(data$OrderDate, label = TRUE) # abbrev labels
```

Bar graph showing product and the sum of sales for each.

```
# Summarize sales by product
product_sales <- data %>%
  group_by(Product) %>%
  summarise(TotalSales = sum(TotalSales, na.rm = TRUE), .groups = 'drop')

# Bar graph
ggplot(product_sales, aes(x = Product, y = TotalSales, fill = Product)) +
  geom_bar(stat = "identity") +
  scale_y_continuous(labels = dollar_format()) +
  labs(title = "Total Sales by Product", x = "Product", y = "Total Sales ($)") +
  theme_minimal() +
  theme(legend.position = "none")
```

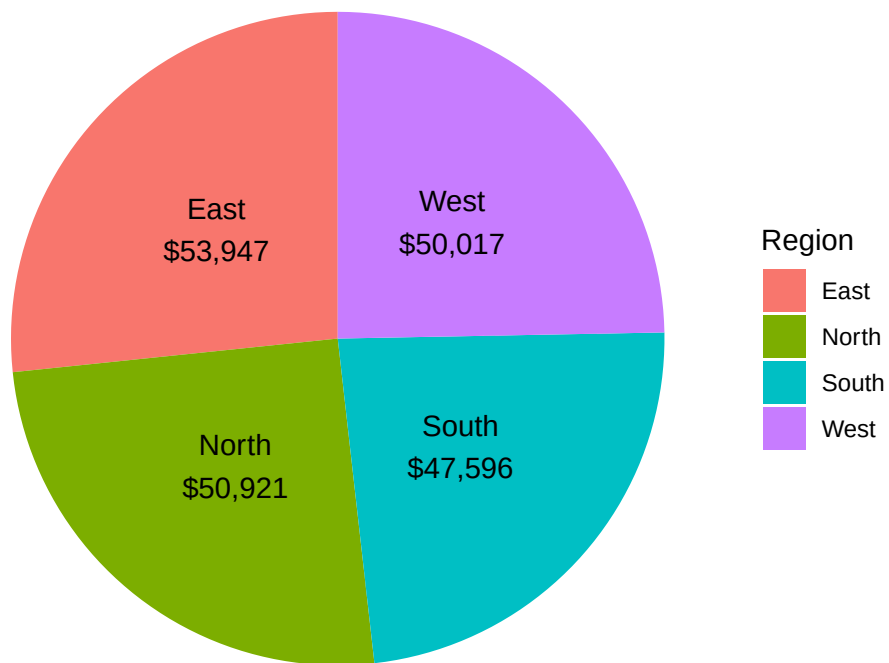


Pie chart showing sum of sales by region.

```
sales_by_region <- data %>%
  group_by(Region) %>%
  summarise(total_sales = sum(TotalSales, na.rm = TRUE)) %>%
  mutate(
    pct = total_sales / sum(total_sales),
    label = paste0(Region, "\n$", scales::comma(total_sales))
  )
# pie chart
ggplot(sales_by_region, aes(x = "", y = total_sales, fill = Region)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y") +
  geom_label(aes(label = label),
    position = position_stack(vjust = 0.5),
    show.legend = FALSE,
    label.size = 0) +
  labs(title = "Sales Distribution by Region") +
  theme_void()
```

```
## Warning: The 'label.size' argument of 'geom_label()' is deprecated as of ggplot2 3.5.0.
## i Please use the 'linewidth' argument instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

Sales Distribution by Region

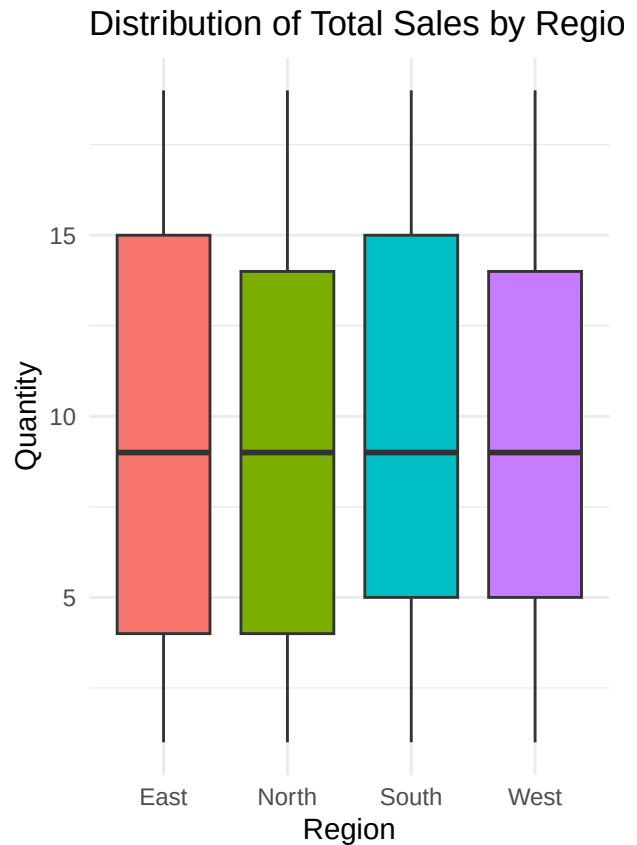
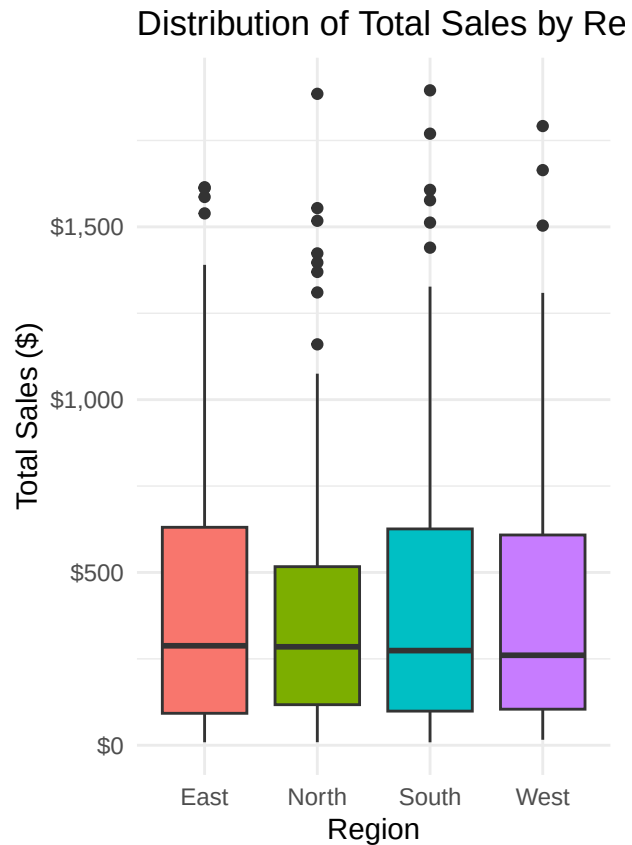


Box Plots - Total Sales and quantity by Region

```
# box plot for total sales by region
p1 <- ggplot(data, aes(x = Region, y = TotalSales, fill = Region)) +
  geom_boxplot() +
  scale_y_continuous(labels = dollar_format()) +
  labs(title = "Distribution of Total Sales by Region", x = "Region", y = "Total Sales ($)") +
  theme_minimal() +
  theme(legend.position = "none")

# box plot for quantity by region
p2 <- ggplot(data, aes(x = Region, y = Quantity, fill = Region)) +
  geom_boxplot() +
  labs(title = "Distribution of Total Sales by Region", x = "Region", y = "Quantity") +
  theme_minimal() +
  theme(legend.position = "none")

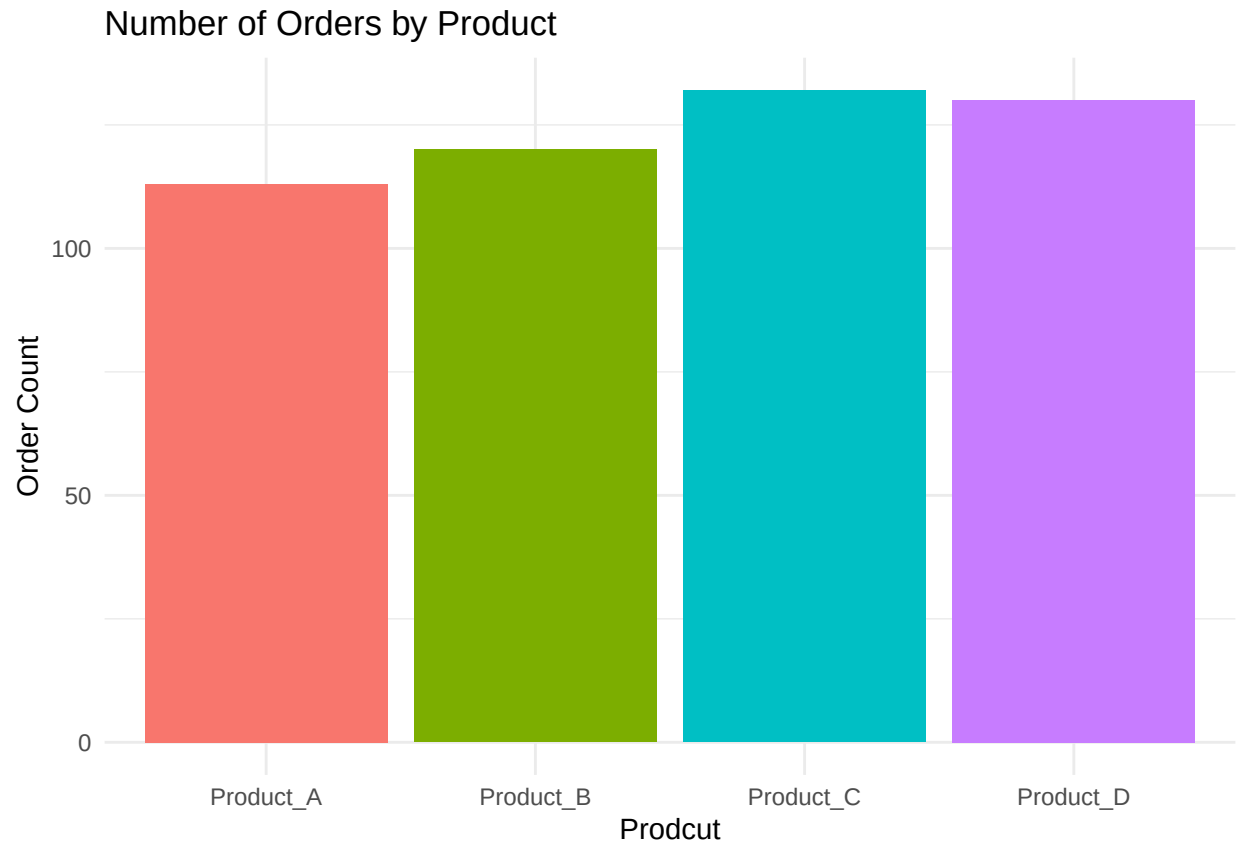
grid.arrange(p1, p2, ncol = 2)
```



Histogram of the count of orders by product

```
# count orders by product
product_count <- data %>%
  group_by(Product) %>%
  summarise(Order_Count = n(), .groups = 'drop')

# bar plot / histogram
ggplot(product_count, aes(x = Product, y = Order_Count, fill = Product)) +
  geom_bar(stat = "identity") +
  labs(title = "Number of Orders by Product", x = "Product", y = "Order Count") +
  theme_minimal() +
  theme(legend.position = "none")
```



A radial plot showing total sales (for all products) by month

```
# Prepare monthly data

monthly_sales <- data %>%
mutate(Month = month(OrderDate, label = TRUE)) %>%
group_by(Month) %>%
summarise(total_sales = sum(TotalSales, na.rm = TRUE))

# Normalize for radar chart

radar_data <- monthly_sales %>%
mutate(total_sales_scaled = total_sales / max(total_sales)) %>%
select(Month, total_sales_scaled)

# Convert to wide format for ggradar

radar_wide <- radar_data %>%
pivot_wider(names_from = Month, values_from = total_sales_scaled)

# ggradar requires a data frame with rownames

radar_df <- as.data.frame(radar_wide)
row.names(radar_df) <- "Monthly Sales"

# Create radar chart
```

```

ggradar(radar_df,
values.radar = c("0","0.5","1"),
grid.min = 0,
grid.mid = 0.5,
grid.max = 1,
legend.position = "none") +
ggtitle("Radial Plot of Total Monthly Sales")

```

Radial Plot of Total Monthly Sales

